Einstein and Machine Learning

The Photoelectric Effect as Statistical Inference

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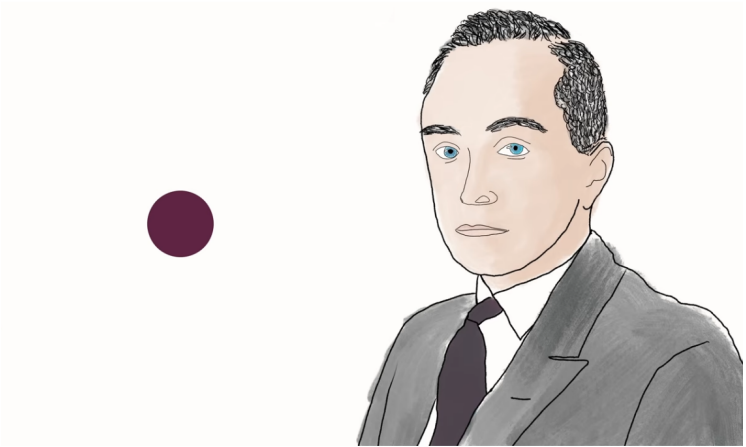
The Born Rule



Weirdest Rule in Physics

Claim

When Einstein proposed the light-quantum hypothesis in 1905 to explain the photoelectric effect, he was doing **machine learning** and **statistical inference**.

1. **Theory** (1905): Einstein proposes a manifold of statistical models (i.e., an "hypothesis class")
2. **Data** (1887–1916): experimenters collect good data
3. **Estimation** (1916): Millikan runs Einstein's regression and recovers Planck's constant to 0.5%

Background: The Photoelectric Effect

The phenomenon (Hertz, 1887; Hallwachs, 1888):

- Shine light on a metal surface; sometimes electrons are ejected
- Apply a *stopping potential* V_0 to halt just the fastest electrons

The puzzle (Lenard, 1902):

- V_0 rises with light *frequency* ν
- V_0 is **independent** of light *intensity* I
- Classical wave theory predicted the opposite

Classical prediction (wrong):

- Wave amplitude $\propto \sqrt{I}$ delivers energy $\Rightarrow$ V_0 should rise with I
- V_0 should be independent of ν

Lenard's Finding as Feature Engineering

Lenard ran an ablation experiment: measured V_0 as a function of both ν and I.

Feature	Effect on V_0	Importance
Frequency ν	Strong $\uparrow$	**High**
Intensity I	None	**Zero**

Conclusion: drop I from the feature set; keep ν.

In modern ML language:

- **Input features:** ν, I, λ, ...
- **Label:** stopping potential V_0
- **Result:** feature-importance analysis eliminates I; selects ν as sole relevant input

Einstein's Model

A light quantum carries energy $E = h\nu$.

A single quantum is absorbed by a single electron. If $h\nu > W$ (work function), the electron escapes.

Energy conservation:

$$eV_0 = h\nu - W$$

Einstein's hypothesis class

$$V_0 = \frac{h}{e}\nu - \frac{W}{e}$$

- Affine in frequency
- Slope $= h/e$ is **universal** across all metals
- Hypothesized before reliable V_0-vs-ν measurements available

Einstein's Inductive Bias

The quantum hypothesis restricts the hypothesis class from *all functions* of $(\nu, I, \lambda, \ldots)$ to a single affine function in ν alone, with a specific universal slope.

Hypothesis class	Size	Data needed
All measurable functions of $(\nu, I, \ldots)$	Infinite-dimensional	$\gg 10^6$
Affine functions of ν alone (Einstein)	2-dimensional	~ 5

Human intelligence (HI) posited a low-parameter functional form.

The Regression Equation

Adding measurement noise and a nuisance parameter:

$$V_{0i} = \underbrace{\alpha}_{\text{intercept}} + \underbrace{\beta}_{h/e} \nu_i + \underbrace{\varepsilon_i}_{\text{noise}}, \qquad \alpha = -\frac{W}{e} + \underbrace{v_c}_{\text{contact EMF}}$$

Parameters of interest:

- $\beta = h/e$ — slope $\Rightarrow$ Planck's h
- α — intercept (metal-specific)

Nuisance parameter: v_c, the contact EMF between emitter and collector.

- Can be as large as 3 V—comparable to V_0 itself
- Shifts intercept; **does not bias slope** if constant within an epoch

Supervised Learning Problem

- **Training data:** $\{(\nu_i, V_{0i})\}_{i=1}^{n}$
- **Task:** estimate $\beta = h/e$
- **Loss:** squared error
- **Hypothesis class:** affine functions of ν
- **Inductive bias:** $\beta > 0$; universal across metals

Computing regression is trivial if you have good data.

29-year gap between Einstein and Millikan comes from data engineering process, not statistical re-specification.

29 Years of Collecting Data (1887–1916)

Experimenter	Year	Data quality	Problem
Hertz / Hallwachs	1887–88	Qualitative only	No labeled (ν, V_0) pairs
Lenard	1902	Qualitative	No reliable numerical V_0
Hughes	1912	3 UV lines	Contact EMF uncontrolled; 65% scatter in h
Richardson & Compton	1912	8 metals	Same problems; 60% scatter
Millikan	**1916**	**Clean surfaces, wide $\Delta\nu$, controlled v_c**	**h to 0.5%**

Millikan's Experimental Design (1916)

Key data-quality innovations ("a machine shop in vacuo"):

- **Fresh surfaces:** electromagnetic knife shaved alkali metal in vacuum—no oxidation, no work-function contamination
- **Spectrally pure lines:** mercury arc with filters to exclude UV companions
- **Wide frequency range:** 5461 Å to 2535 Å ($\Delta\nu = 6.3 \times 10^{14}$ Hz, nearly a factor of two)
- **Controlled nuisance:** Kelvin double-balance measured v_c continuously; stable to ± 0.01 V over 9 hours

Having a wide frequency range helps (Cramér–Rao bound):

$$\mathrm{SE}(\hat{\beta}) = \frac{\sigma}{\sqrt{S_{\nu\nu}}}, \quad S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2$$

Wider $\Delta\nu \Rightarrow$ larger $S_{\nu\nu} \Rightarrow$ smaller SE.

Millikan's Data

λ (Å)	ν (10^{14} Hz)	V_0^{Na} (V)	V_0^{Li} (V)
5461.0	5.490	0.454	—
4339.4	6.908	1.039	0.499
4046.8	7.409	1.246	0.706
3650.2	8.213	1.578	1.037
3125.5	9.591	2.147	1.606
2534.7	11.827	3.030	2.529

True stopping potential $= V_{0,\mathrm{obs}} + v_c$ (Na: $v_c = 2.51\,\mathrm{V}$; Li: $v_c = 1.52\,\mathrm{V}$)

- 6 labeled training examples per metal; two metals
- Data span nearly a factor of two in frequency

Estimating Planck's Constant

Primary regression: sodium, five points

Standard OLS formula:

$$\hat{\beta} = \frac{S_{\nu V}}{S_{\nu\nu}} = \frac{3.830}{9.277} = \mathbf{4.129 \times 10^{-15}}\ \mathrm{V{\cdot}s}$$

	Value	
R^2	≈ 0.99999	
$\mathrm{SE}(\hat{\beta})$	4.6×10^{-18} V·s	
Implied h (Millikan's e, 1916)	6.573×10^{-34} J·s	-0.80%
Implied h (modern NIST e)	6.614×10^{-34} J·s	-0.18%
Modern exact h	6.626×10^{-34} J·s	

h recovered to **0.5%** of its modern value from **5 data points**.

MLE = OLS Under Gaussian Errors

Parametric model:

$$V_{0i} = \alpha + \beta\,\nu_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

Log-likelihood:

$$\ell(\alpha, \beta, \sigma) = -\frac{n}{2}\ln(2\pi) - n\ln\sigma - \frac{1}{2\sigma^2}\sum_i (V_{0i} - \alpha - \beta\nu_i)^2$$

Maximising ℓ over (α, β) minimises RSS—exactly OLS:

$$(\hat{\alpha}_{\text{MLE}}, \hat{\beta}_{\text{MLE}}) = (\hat{\alpha}_{\text{OLS}}, \hat{\beta}_{\text{OLS}})$$

The Cramér–Rao Bound and Experimental Design

Fisher information and Cramér–Rao bound for $\beta = h/e$:

$$\mathcal{I}_{\beta\beta} = \frac{S_{\nu\nu}}{\sigma^2}, \qquad \mathrm{Var}(\tilde{\beta}) \geq \frac{\sigma^2}{S_{\nu\nu}}$$

OLS achieves this bound (Gauss–Markov + Cramér–Rao).

Dataset	n	$S_{\nu\nu}$ (10^{28} Hz^2)	SE($\hat{\beta}$) (10^{-18} V·s)
Restricted range (like Hughes)	3	1.96	10.0
Millikan 5-pt	5	9.28	4.6
Millikan full 6-pt	6	28.14	2.6

Design lesson: span frequencies widely.
Hughes's narrow range gives a standard error **2.2**$\times$ **larger** than Millikan's full range.

Universality: Zero-Shot Transfer Across Metals

The quantum hypothesis predicts $\beta = h/e$ is the same for *all* metals.

Separate regressions:

$$\hat{\beta}_{\mathrm{Na}} = 4.129 \times 10^{-15}\ \mathrm{V{\cdot}s}$$
$$\hat{\beta}_{\mathrm{Li}} = 4.137 \times 10^{-15}\ \mathrm{V{\cdot}s}$$
$$\text{difference} = 0.19\%$$

Wald test of $H_0 : \beta_{\mathrm{Na}} = \beta_{\mathrm{Li}}$: $z \approx 0.87$ ($\ll 1.96$). Cannot reject universality.

In ML language:

- **Pre-training:** fit slope on sodium
- **Zero-shot transfer:** apply slope to lithium
- **Fine-tuning needed:** only the intercept
- **Transfer error:** 0.2%

Historical Estimates of Planck's Constant

Authors	Year	h (10^{-27} erg·s)	Error
Hughes	1912	3.55–5.85	−12 to −46%
Richardson & Compton	1912	3.5–5.8	−12 to −47%
Planck (black body)	1913	6.415	−3.2%
Millikan (Na, 5-pt)	1916	6.573	−0.80%
Millikan (Na, 9-pt)	1916	6.578	−0.73%
Millikan (Li)	1916	6.584	−0.64%
Millikan mean	**1916**	**6.579**	**−0.72%**
Modern NIST 2022	2022	6.6261	0

Pre-Millikan estimates: biased downward 10–47%.
Three sources: stray UV, narrow $\Delta\nu$, uncontrolled v_c.
Millikan eliminated all three.

Millikan's Doubts

Millikan confirmed Einstein's equation to 0.5%. Then wrote:

> *"Despite then the apparently complete success of the Einstein equation, the physical theory of which it was designed to be the symbolic expression is found* **so untenable** *that Einstein himself, I believe, no longer holds to it."*
>
> Millikan (1916), p. 388

He had hoped the experiment would **refute** the equation.

- Confirming a regression equation says little about the physical structure that generated it
- Compton scattering (1923) finally convinced him
- Millikan's 1923 Nobel prize: "...first direct experimental proof ...of the complete validity of the Einstein equation"

ML-to-Physics Dictionary

Scientific step	**ML name**	**Who**
Propose $E = h\nu$; derive $V_0 = (h/e)\nu - W/e$	Inductive bias + hypothesis class	Einstein (1905)
V_0 depends on ν not I	Feature engineering	Lenard (1902)
Measure (ν_i, V_{0i}) pairs	Labeled training data	Millikan (1916)
Control contact EMF v_c	Nuisance-variable identification	Millikan (1916)
Min. $\sum(V_{0i} - \hat{V}_{0i})^2$	MSE training (OLS/MLE)	Millikan (1916)
$\lvert\hat{\beta} - h/e\rvert < 1\%$	Test accuracy	Millikan (1916)
Na and Li give same $\hat{\beta}$	Zero-shot transfer learning	Millikan (1916)

The Hypothesis Class

Einstein's hypothesis class:

$$\mathcal{H} = \left\{ f_{\alpha,\beta} : \nu \mapsto \alpha + \beta\nu \;\middle|\; \alpha \in \mathbb{R},\ \beta > 0 \right\}$$

with the cross-metal restriction that β is universal.

Properties:

- VC dimension $= 2$
- One-parameter family of parallel lines
- Deduced from $E = h\nu$
- Five training points are more than sufficient

What a DL ML system misses:

1. Does not identify $\hat{\beta} = 4.129 \times 10^{-15}$ as h/e
2. Does not impose $\beta_{\mathrm{Na}} = \beta_{\mathrm{Li}}$ from first principles
3. Does not decompose intercept into W/e vs. v_c
4. Does not reduce data requirement from $O(10^3)$ to 5

Einstein vs. a Deep Neural Network

Epoch	n	Einstein model	DNN (VC $\approx$ 5700)	DNN outcome
Lenard 1902	≈ 0	Sufficient*	Insufficient	Cannot train
Hughes 1912	~ 30	Sufficient	Insufficient ($\times$190)	Memorises; no gen.
Millikan 1916	5	Sufficient	Insuf. ($\times$1140)	Interpolates only
Modern era	~ 5000	Sufficient	Sufficient	Recovers line; cannot identify h/e

*Functional form from theory; only two free parameters remain.

A three-layer DNN (16 neurons/layer): $p \approx 593$ parameters, VC dim ≈ 5700. Reliable 1% generalisation requires $n \gtrsim 5700$—short by three orders of magnitude in 1916.

The DNN on Millikan's Five Points

What happens if you train any DNN on $n = 5$ data points?

Interpolation regime (any DNN with $p > 5$):

- Training loss $= 0$ (memorises all points)
- Function is unconstrained between points
- Cannot estimate slope uncertainty: zero residual df

OLS with $n = 5$, $p = 2$:

- 3 residual degrees of freedom
- $\mathrm{SE}(\hat{\beta}) = 4.6 \times 10^{-18}$ V·s
- 0.11% relative precision, at the Cramér–Rao bound

Einstein's theory made five training points sufficient (not more data, not a better algorithm)

DNNs win for problems without a known parsimonious parametric family (images, proteins, language). The photoelectric problem sits at the *opposite* extreme.

The Hypothetico-Deductive Loop

Step	Science	ML
Propose $E = h\nu$	Theoretical hypothesis	Model architecture
Derive $V_0 = (h/e)\nu - W/e$	Deduction	Hypothesis class
Choose ν, not I	Experiment design	Feature engineering
Measure (ν_i, V_{0i})	Observation	Training data
Minimise RSS	Least squares	MSE training
$\lvert\hat{\beta} - h/e\rvert < 1\%$	Confirmation	Test accuracy
Na and Li: same $\hat{\beta}$	Universality test	Zero-shot transfer

Modern AI systems *induce* a linear law from lots of data and search over many parameters of a DNL.
Einstein's HI system deduced an affine law (no data?)

Unintended Consequence: Born's Rule

Einstein's quantum hypothesis required $|E|^2 \propto$ photon density.
The classical wave intensity $|E|^2$ becomes a probability density. Einstein called this the *Gespensterfeld* (ghost field).

Max Born (1926) formalised the same logic for the Schrödinger wave function:

$$\boxed{|\psi|^2 = \text{probability density}}$$

Two Nobel Prizes:

- 1921: Einstein — for the equation that made the measurement possible
- 1954: Born — for the probabilistic interpretation the equation made *inevitable*

Einstein spent the rest of his life rejecting Born's rule. He won the Nobel for a paper whose logic seeded the interpretation he most opposed.

Summary and Conclusions

The Thesis, Revisited

Einstein was doing machine learning in 1905—but the most important step was the one no ML system has yet automated: **specifying the hypothesis class**.

Three principles:

1. **Theoretical priors guide design.** $E = h\nu$ fixes the functional form; only 5 training points suffice for 0.1% precision.
2. **Data quality is the binding constraint.** The regression is trivial; assembling good data took 29 years.
3. **Confirming an equation $\neq$ endorsing the model.** Millikan confirmed the slope to 0.5% while rejecting the light-quantum interpretation.

A data-driven system given only (ν_i, V_{0i}) pairs can recover $\hat{\beta} = 4.129 \times 10^{-15}$ V·s but cannot know it is h/e. **Physical meaning requires physical theory.**

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